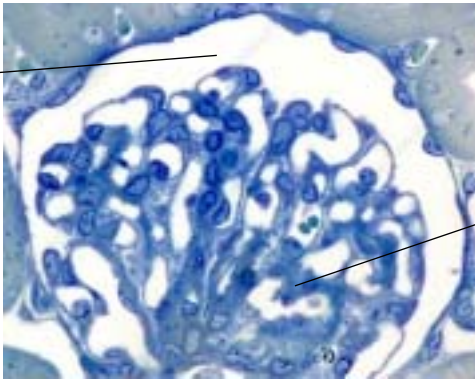


6(a)	 A micrograph showing a cross-section of a placental villus. Label 'F' points to the fetal blood vessel (artery) within the villus. Label 'P' points to the maternal blood vessel (sinus) surrounding the villus.	2
6(b)(i)	$\frac{118 - 52}{60} \text{ or } \frac{118 - 52}{69} \text{ or } \frac{118 - 52}{51} ;$ 1.1 or 1.0 or 1.3 ; $\text{cm}^3 \text{ min}^{-1} \text{ y}^{-1}$; A $\text{cm}^3 \text{ min}^{-1} / \text{year}$ A year(s)^{-1} R yr or yrs or mins	3

Question	Answer	Marks
6(b)(ii)	any three from: 1 high(er), hydrostatic / blood, pressure causes higher GFR ; ora 2 water / solute, potential gradient between glomerulus and Bowman's capsule affects GFR ; 3 kidney disease / named, decreases GFR ; e.g. damage to nephron or basement membrane / cancer 4 dehydration / lower water potential of blood, decreases GFR ; ora 5 high altitude decreases GFR ; ora 6 drugs / named, affects GFR ; e.g. diuretics / beta blockers / vasodilators / NSAIDS / ACE inhibitors 7 larger difference between afferent arteriole lumen and efferent causes higher GFR ; ora	3

Question	Answer	Marks
6(c)	any seven from: 1 Na⁺, actively transported / pumped, out of cells ; 2 Na⁺ enter blood ; 3 Na⁺ concentration in cells decreases or Na⁺ concentration gradient (between pct lumen and cytoplasm) / described ; 4 Na⁺ enter cells from, filtrate / lumen ; 5 by facilitated diffusion ; 6 cotransport of, glucose / amino acids ; 1 secondary active transport 7 water potential of cells decreases ; 8 water enters cells from (filtrate / lumen) by, osmosis / down the water potential gradient ; 9 glucose / amino acids, leave cells by, (facilitated) diffusion / described ; 10 glucose / amino acids, reabsorbed or move into , tissue fluid / blood ; 11 AVP ; e.g. other named ions reabsorbed	7